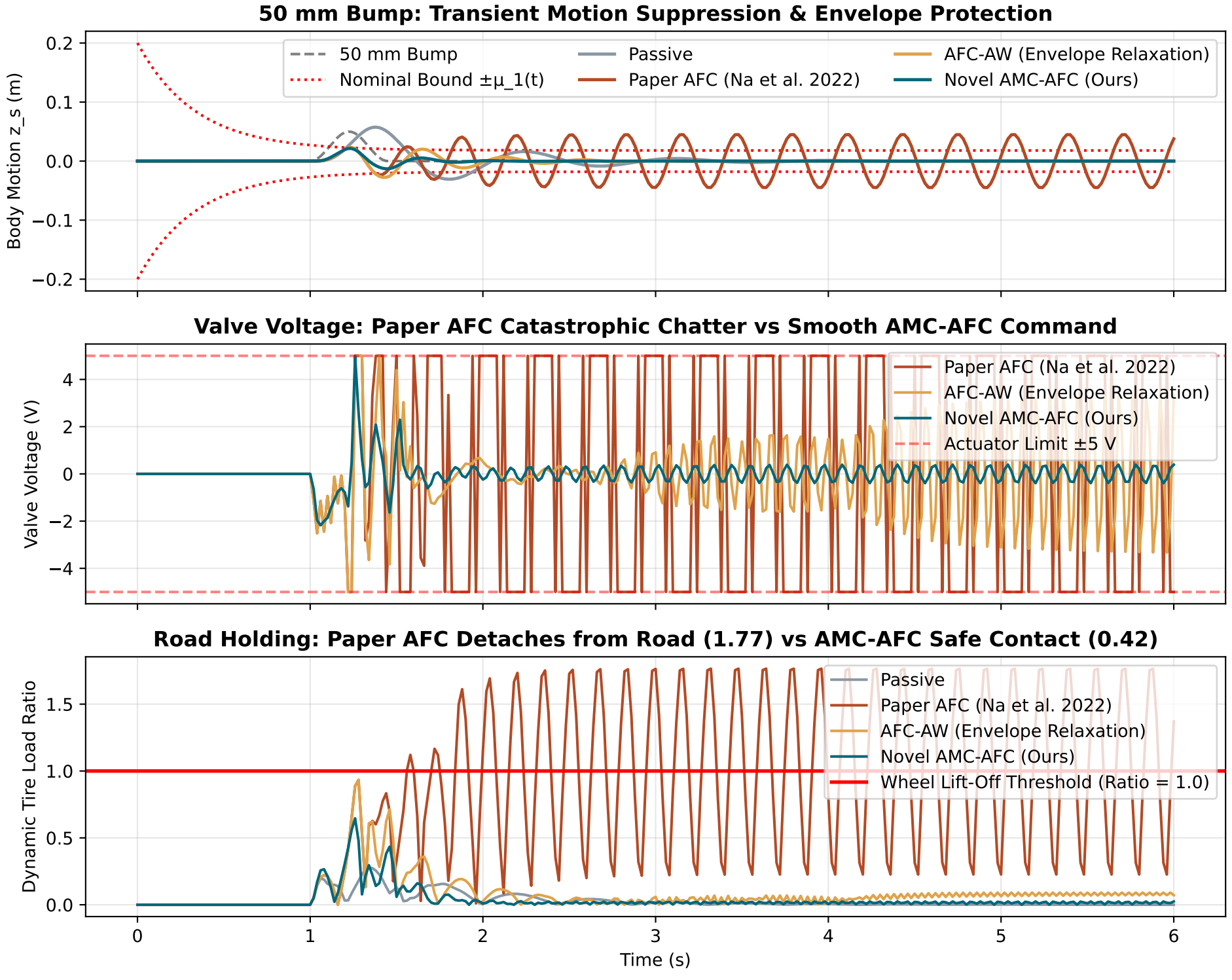
50 mm Bump: Transient Motion Suppression & Envelope Protection
Body Motion z_s (m)
0.2
0.1
0.0
−0.1
−0.2
50 mm Bump
Nominal Bound ±μ_1(t)
Passive
Paper AFC (Na et al. 2022)
AFC-AW (Envelope Relaxation)
Novel AMC-AFC (Ours)
Valve Voltage: Paper AFC Catastrophic Chatter vs Smooth AMC-AFC Command
Valve Voltage (V)
4
2
0
−2
−4
Paper AFC (Na et al. 2022)
AFC-AW (Envelope Relaxation)
Novel AMC-AFC (Ours)
Actuator Limit ±5 V
Road Holding: Paper AFC Detaches from Road (1.77) vs AMC-AFC Safe Contact (0.42)
Dynamic Tire Load Ratio
1.5
1.0
0.5
0.0
Passive
Paper AFC (Na et al. 2022)
AFC-AW (Envelope Relaxation)
Novel AMC-AFC (Ours)
Wheel Lift-Off Threshold (Ratio = 1.0)
0
1
2
3
4
5
6
Time (s)